Wei Fu

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Research Interests

- Safe and stable learning-based control of dynamical systems
- Neural operators and physics-informed machine learning
- Multi-agent systems and risk estimation
- Optimization and control theory

Education

University of Southern California Los Angeles, CA
M.S. in Electrical Engineering, Viterbi School of Engineering *08/2025 – 05/2027 (expected)*

Northeastern University Shenyang, China
B.Eng. in Automation | GPA: 87.57/100 *09/2021 – 06/2025*

Publications & Preprints

[1] Gengshuo Liu*, **Wei Fu***, Italo Ivo Lima Dias Pinto, Yorie Nakahira, Paul Bogdan. "TF-Risk: Estimating False-Consensus Risk in Multi-Agent Systems with Time-Fractional Dynamics." *Under review, AAAI 2027.* (*Equal contribution)

[2] Gengshuo Liu, Shixuan Li, Xiongye Xiao, **Wei Fu**, Yucheng Zhao, Paul Bogdan. "Lyapunov-Inspired Neural Operators for Stable Long-Rollout PDE Prediction." *Preprint.*

[3] **Wei Fu**. "Exploratory Data Analysis and Machine Learning Models for Stroke Prediction." *Int'l Conf. on Data Analysis and Machine Learning (DAML), 2023.* DOI: 10.5220/0000185700003885

Research Experience

University of Southern California 02/2026 – Present
Graduate Researcher, advised by Prof. Paul Bogdan

- Co-first-authored TF-Risk [1], a framework for estimating false-consensus risk in multi-agent systems governed by time-fractional dynamics, capturing long-memory effects missed by integer-order consensus models; designed and implemented the risk-estimation pipeline and led the simulation studies validating the framework across consensus scenarios. Collaboration with Prof. Yorie Nakahira (CMU).
- Contributed to Lyapunov-inspired neural operators [2] that embed stability certificates into operator learning; implemented neural-operator baselines and ran long-rollout experiments on benchmark PDEs, demonstrating improved rollout stability over standard neural-operator methods.

Digital Twin-Based Product Quality Prediction in Industrial Processes 11/2024 – 06/2025
Undergraduate Thesis, advised by Prof. Minghai Jiao & Chang Liu, Northeastern University

- Built physics-informed quality-prediction models for cold-rolling and annealing processes, injecting features derived from thermodynamic/kinetic models (e.g., grain size, dislocation density) into XGBoost, CatBoost, and TabTransformer predictors.
- Optimized hyperparameters and feature selection with evolutionary algorithms; developed an interactive Unity digital-twin interface for real-time process monitoring and mechanical-property prediction.

Pin-QSSVM: Quadratic-Surface SVM with Pinball Loss for Robust Classification 06/2023 – 08/2024
Research Assistant, advised by Asst. Prof. Zheming Gao, Northeastern University

- Proposed Pin-QSSVM, integrating the pinball loss into the quadratic-surface SVM framework for more flexible error penalization on noisy and imbalanced datasets.
- Solved the model via stochastic gradient descent and second-order cone programming; achieved +11% accuracy, −9% misclassification error, and +14% F1 over standard QSSVM on public benchmarks and proprietary healthcare/finance data.

Machine Learning for Stroke Prediction 01/2023 – 09/2023
Independent Project, advised by Prof. Mark Vogelsberger; published at DAML 2023 [3]

- Benchmarked logistic regression, random forest, and XGBoost on clinical stroke data; best model (random forest) reached 94.1% accuracy and 0.97 F1, with analysis of precision–recall trade-offs across models.

Intelligent Monitoring System for School-Bullying Detection 01/2022 – 09/2023
Team Leader, advised by Prof. Dongyue Chen, Northeastern University

- Built a real-time multi-target tracking pipeline (YOLOv5 + DeepSORT) to detect harmful interactions in crowded school environments, trained on a public violence dataset augmented with self-collected data; deployed across heterogeneous school camera systems with real-time alerts.

Industry Experience

Chongqing Jifeng Geological Exploration Co., Ltd. 06/2022 – 09/2022
Data Engineer Intern

- Cleaned and preprocessed geological mapping data (remote sensing, seismic, well logs); trained random-forest and SVM classifiers for lithology classification to support mineral-resource zoning.

Selected Projects

Edge-Computing Control System for Motor Speed Regulation 05/2024 – 07/2024

- Implemented PID and fuzzy self-adaptive PI controllers on an EdgeX Foundry/Docker edge stack with MQTT communication; integrated and calibrated microcontroller, rectifier, DC motor, and speed-sensor hardware, with custom Go plugins and a real-time monitoring UI.

Honors & Awards

- Outstanding Student Award, Northeastern University, 2023
- First Prize, Liaoning Mathematics Competition, 2022

Skills

Programming: Python (PyTorch, scikit-learn), MATLAB/Simulink, C++, Go
Tools: Docker, EdgeX Foundry, Unity, Git